

## Evaluation of Response Time Performance in Medication Reminder App: A Comparative Study Between iOS and Android Applications.

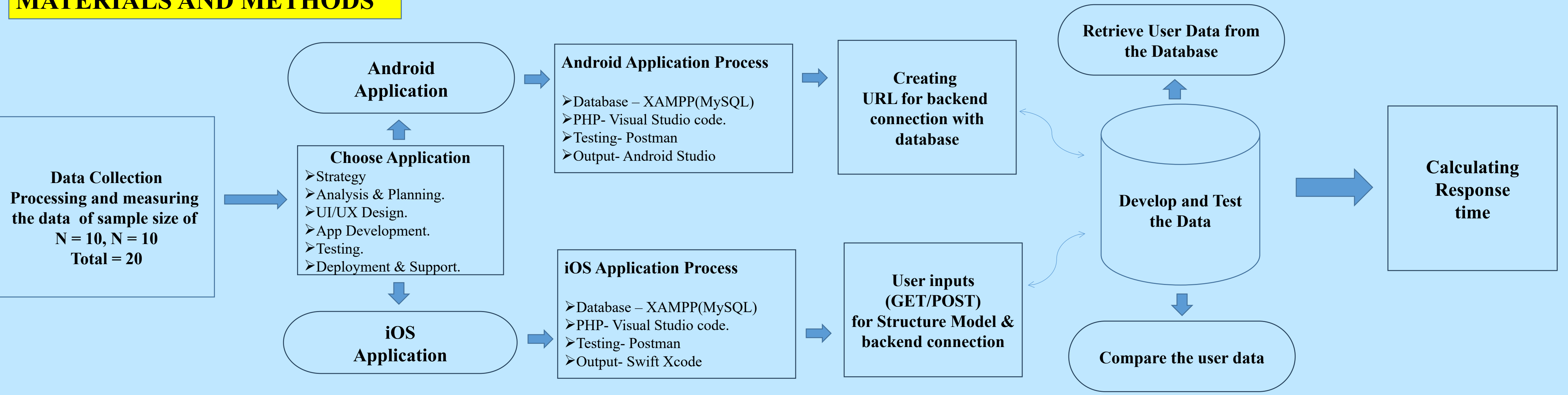
### INTRODUCTION

- The Medication-Reminder is a comprehensive solution that uses voice app notification to remind patients to take their medications at the appointed times.
- It enables smooth communication and effective progress tracking between medical professionals and patients. It improves the healing process for patients using tablets by offering precise scheduling and thorough progress tracking.
- The aim of this project is to assess and contrast the navigation response times and user navigation durations experienced during the treatment process on both the Medication-Reminder Android application and the iOS platform.
- We found that the iOS application had a faster navigation reaction time than the Android version when we evaluated which improves both user experience and operational effectiveness.
- Therefore, this study serves as a critical step towards leveraging technology to enhance patient care and promote better health management through mobile medication-Reminder applications.



Fig 1:Medication-Reminder

### MATERIALS AND METHODS



### RESULTS

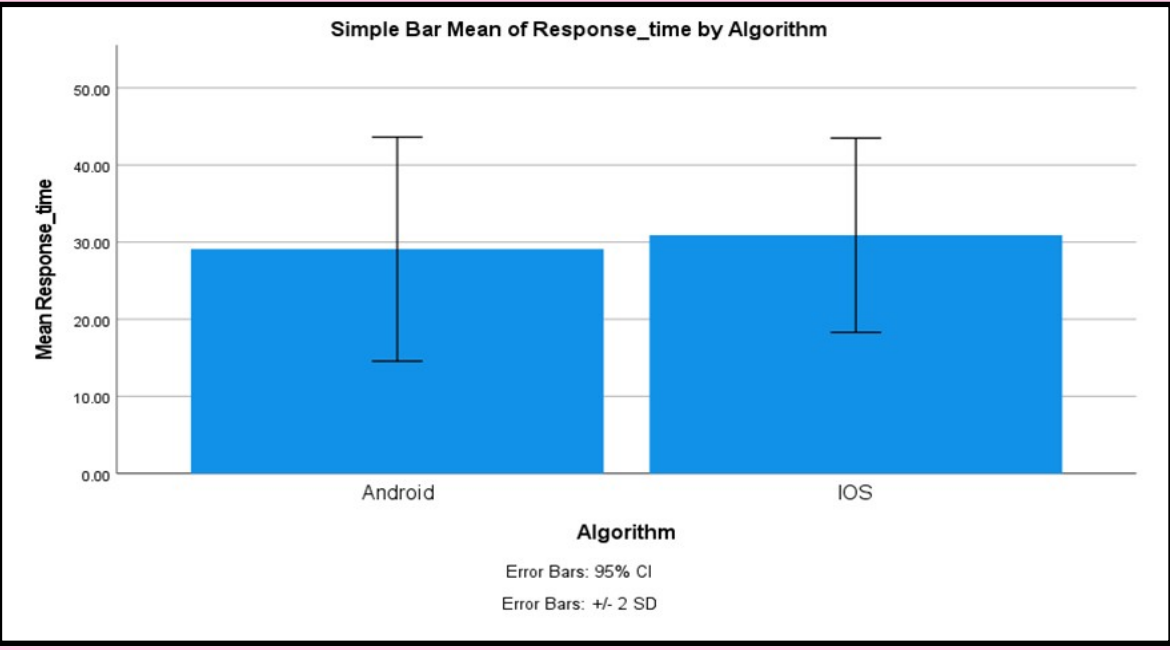


Fig 2: Comparision of Andriod and iOS Application in terms of Response Time.

| Group Statistics |           |    |         |                |                 |
|------------------|-----------|----|---------|----------------|-----------------|
|                  | Algorithm | N  | Mean    | Std. Deviation | Std. Error Mean |
| Response time    | Android   | 10 | 29.1000 | 7.26407        | 2.29710         |
|                  | iOS       | 10 | 30.9000 | 6.29727        | 1.99137         |

- Comparision of Andriod and iOS Application in terms of Response Time. The Mean Recovery Rate of iOS is significantly higher than Android in Medication-Reminder.
- UI responsiveness significantly impacts user satisfaction and perception of application performance.
- If users experience delays or lag in the interface responsiveness, it can lead to frustration and dissatisfaction, regardless of the actual response time of the application.

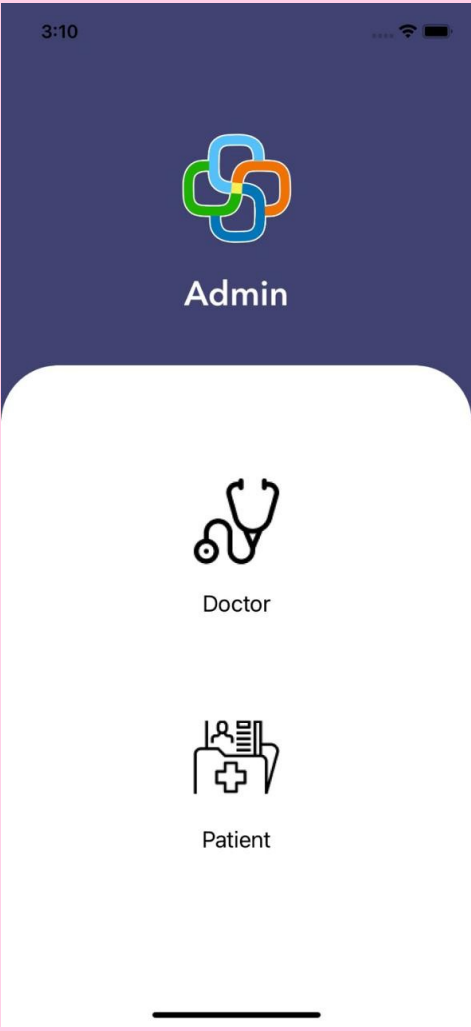


Fig 3:Admin Page

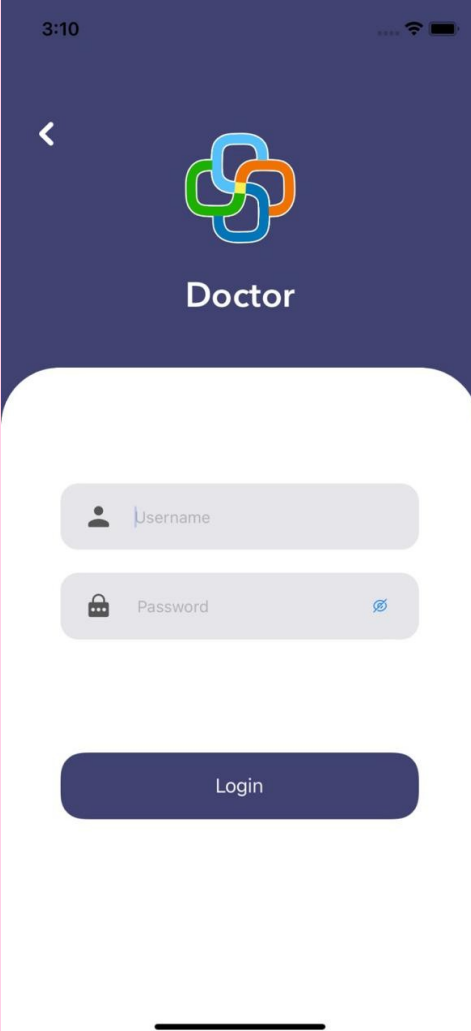


Fig 4:Doctor Login

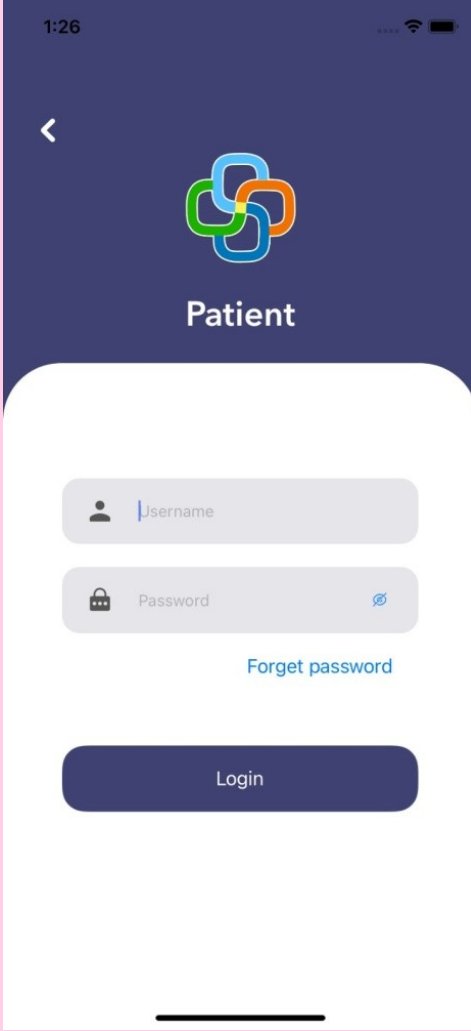


Fig 5:Patient Login

### DISCUSSION AND CONCLUSION

- Based on T-test Statistical analysis, the significance value of p=0.007 (independent sample T - test p<0.05) is obtained and shows that there is a statistical significant difference between the group 1(iOS Application) and group 2(Android Application).
- Overall, the Response time of the Medication-Reminder iOS Application is 29.33ms and it is better than the Android Application. The Android Application is 34.58ms.
- From the work, it is concluded that the iOS Application attains the speed response time when comparing with Android Application in Medication-Reminder
- Based on the statistical analysis and comparison of response times, prioritizing the optimization of the iOS version of Medication-Reminder is recommended to maintain its superior speed performance over the Android version, ensuring a seamless user experience.
- This superior user experience on iOS could lead to higher user satisfaction ratings, increased retention rates, and potentially greater word-of-mouth recommendations, ultimately contributing to the overall success and popularity of the application.

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